

Predicting Monthly Passenger Air Traffic in Brazil

Big Data Project - TEAM 12

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Aviation Market Traffic: Seasonality and Structural Breaks



Annual Seasonality
Seasonal decomposition reveals a clear, stable annual pattern with predictable peaks



COVID-19 Shock
A sharp decline occurred in early 2020, with international air traffic collapsing almost



Domestic Lead
Domestic traffic recovered faster and exceeded pre-pandemic levels earlier than international routes.

01.

Background

Brazil is one of the largest aviation markets in the world. Future forecasts are essential to anticipate capacity constraints, plan airports expansions and resources allocation, and monitor economy.

02.

Objective

Develop a time-series forecasting model that predicts passenger air traffic in Brazil.

03.

Dataset

[Brazil Air Traffic Data 2000–2025 \(ANAC\)](#)

Project overview

01.

Actuality

Dataset contains record up to 2026. Also, in its description is claims to be expected to update as new portions of data arrive.

This makes our project useful both now and in nearest future.

02.

Reliability

Dataset contains 20M+ records. It allows us produce generalizable insights.

03.

Usability

Air traffic in general is really profitable field of business. We theoretically can generate something useful for business

Dataset overview

Project pipeline

Framework



Data Preparation

Data:

- downloaded
- preprocessed by selecting relevant columns and converting data types
- stored in PostgreSQL
- exported to HDFS using Sqoop for distributed processing
- multiple storage formats (Text, Avro, Parquet) are benchmarked by write/read time, and storage size, with parquet + gzip selected

EDA

- Loaded dataset into Hive.
- applied optimizations for further steps.
- Engineered queries.
- Created charts with useful insights based on them.

Data Modeling

Data Cleaning and Aggregation

- filter flights by type (must be regular and passenger), number of passengers ($\leq \sim 900$), year (≤ 2025)
- drop nan values
- aggregate passengers by month

Feature Engineering

- CyclicalMonthEncoder

$$\sin\left(2\pi \cdot \frac{\text{month}}{12}\right), \quad \cos\left(2\pi \cdot \frac{\text{month}}{12}\right)$$

FEATURE_COLUMNS:

```
"year",  
"time_index",  
"quarter",  
"is_filled",  
"month_enc_sin",  
"month_enc_cos",  
"passengers_lag_1",  
"passengers_lag_12",  
"passengers_lag_24",  
"passengers_roll_mean_3",  
"passengers_roll_mean_12",  
"passengers_roll_mean_24",  
"passengers_roll_std_12"
```

Models, hyperparameters

- SparkXGBRegressor (`max_depth`, `min_child_weight`, `subsample`)
- GBTRRegressor (`maxDepth`, `minInstancesPerNode`, `subsamplingRate`)

Cross-validation



Evaluation

- RMSE
- R^2

Dashboards presentation

Results discussion

Success and limitations



The preprocessing is well-structured with 13 meaningful features capturing both operational and contextual information



Multiple storage formats (Text, Avro, Parquet) with different compression methods were systematically tested, demonstrating a solid experimental approach.



Partitioning by year was introduced during optimization, improving query performance and data organization in Hive



Incorrect train/test split -> poor results

Challenges

- choosing core features at the initial stage
- Interpretation of data statistics for business tasks
- model selection and fine tuning of parameters

Q&A